

EVIDENCE REVIEW

How Leading Organizations Assess Foundational Learning

What their grading practices mean for Sabi's
foundational literacy and numeracy architecture

Pratham / ASER • TaRL • J-PAL • UNICEF • USAID / RTI •
World Bank • UNESCO • Nigeria NALABE • Save the Children • Room to Read

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A practical assessment architecture—not a literature review alone

Executive conclusion

Leading foundational learning organizations generally do **not** use one percentage grade to decide whether a child is learning. They combine several different instruments, each with a different job:

1. **Placement:** a short, progressive assessment identifies the highest level a child can comfortably perform and places teaching at that level.
2. **In-lesson formative evidence:** each response tells the teacher what to affirm, reteach, or scaffold immediately.
3. **Mastery and progress monitoring:** repeated, independent performance across different items determines whether a skill is secure and whether the child should advance.
4. **Benchmark or system reporting:** a separate, standardized assessment measures change over time and supports external reporting.

That separation is the central recommendation for Sabi. A single conversational turn should never simultaneously function as encouragement, a permanent grade, a TaRL placement decision, and pilot-impact evidence.

The strongest common practices are:

- assess observable skills rather than age, grade, confidence, or “general ability”;
- use progressive tasks and record the highest stable level;
- use deterministic item keys and explicit scoring rules;
- keep accuracy, fluency, comprehension, and independence separate;
- use frequent formative checks to change the next teaching move;
- treat a correct answer after a scaffold as evidence that the scaffold worked, not as independent mastery;
- adapt language, content, examples, and benchmarks to the local context;
- distinguish “wrong” from “not scorable”; and
- validate automated scoring against trained human judgment before using it for consequential decisions.

Recommended decision

Build Sabi as a **four-layer assessment system**, not an LLM-generated grade:

Layer	Question it answers	Sabi output
Turn evidence	What happened on this response?	Scorable/not scorable; correct/incorrect; independent/self-corrected/scaffolded
Skill mastery	Is this specific skill secure?	Not introduced, emerging, developing, secure, retained
Placement	What should Sabi teach next?	Current literacy and numeracy level, with prerequisite gaps
Evaluation	Is the programme improving learning?	Standardized baseline/progress/endline results, kept separate from tutoring data

For the current numeracy-first pilot, Sabi should count a skill as **secure** only after at least **four of the last five scorable, independent probe items are correct**, across at least two calls and more than one item form. This is a proposed Sabi rule derived from ASER’s 4-of-5 task rules and the 80–90% mastery range used in mastery-learning literature; it is **not** a universal NGO standard and must be validated during the pilot.

1. What “grading” means in foundational learning

The organizations reviewed use assessment for different purposes. Confusion arises when a rule designed for one purpose is reused for another.

1.1 Placement is not a report-card grade

Pratham’s ASER and Teaching at the Right Level (TaRL) use simple, oral, one-to-one tools to find the highest reading or mathematics level a child can demonstrate. The result is an actionable category—such as beginner, letter, word, paragraph, or story—not a percentage mark. TaRL then groups and teaches children by current level rather than by age or school grade. Children are reassessed and move as their learning changes. ^{[1][2][3]}

1.2 Formative assessment is a teaching action

UNICEF defines formative assessment by what happens next: evidence must inform instruction. Its guidance distinguishes beginning-of-term screening, daily mastery monitoring, progress monitoring weekly or every two to four weeks for learners below goals, and mid/end-year benchmark assessment. ^[6]

The World Bank makes the same distinction. Classroom assessment should diagnose, provide continuous feedback, and adjust instruction; large-scale assessment answers a different system-level question. In Sobral, Brazil, continuous classroom checks coexist with twice-yearly census assessments and individualized progress reports. ^{[10][11]}

1.3 Benchmark proficiency is not daily mastery

UNICEF’s MICS Foundational Learning Skills indicator classifies a child as having foundational reading skills only if the child reads at least 90% of a story correctly and answers all three literal and both inferential questions correctly. Foundational numeracy requires success on every question across number reading, number discrimination, addition, and pattern recognition. This deliberately strict rule supports population reporting; it is not a recommended daily tutoring threshold. ^[7]

UNESCO’s Global Proficiency Framework and Nigeria’s NALABE use performance bands—below partially meets, partially meets, meets, and exceeds. NALABE’s cut scores differ by subject and grade because experts set standards against the skills represented by each test. For example, its P3 “meets” cut is 63% in mathematics and 70% in English, while the “exceeds” cuts are 87% and 97%, respectively. This is strong evidence against imposing one universal percentage across all Sabi skills. ^{[12][13]}

2. How leading organizations score and use results

2.1 Pratham / ASER: coarse, transparent level classification

ASER assesses children aged 5–16 with the same progressive foundational tool, irrespective of school grade. It records the highest level the child can comfortably reach. ^[1]

Reading rules

- letter level: at least four of five letters correctly recognized;
- word level: at least four of five words correctly read;
- paragraph level: reads a Grade 1-level paragraph like connected sentences, with no more than three mistakes; and
- story level: reads a Grade 2-level story fluently and comfortably, with no more than three mistakes. ^[4]

Arithmetic rules

- number recognition: at least four of five numbers correct;
- subtraction with borrowing: both of two problems must be correct to advance;
- division: quotient and remainder must both be correct; and
- an apparent careless error can receive another attempt, but the recorded result is still the highest demonstrated level. ^[5]

Lesson for Sabi: transparent pass rules and an interpretable level are more useful for placement than an opaque composite score. However, ASER’s small number of items is designed for quick classification, not fine-grained mastery certification.

2.2 TaRL and J-PAL: assess, group, teach, reassess

TaRL adapts the ASER-style approach to instruction. Its assessment is quick, oral, one-to-one, progressive, and adapted to the local language and context. Reading commonly progresses from letters or symbols to everyday words, a Grade 1 paragraph, a Grade 2 story, and comprehension. Mathematics commonly covers number recognition and basic operations. The information is used to group learners, set goals, track progress, and prioritize support. ^{[2][3]}

J-PAL’s evidence synthesis identifies six randomized evaluations across implementation models. The effective model is not merely “test children”; it aligns assessment, level-based grouping, appropriate activities, mentoring, and review. ^[14]

UNICEF’s Nigeria TaRL guide recommends at least baseline, midline, and endline assessment while allowing movement between groups as teachers observe progress. Nigeria’s implementation reported substantial baseline-to-endline increases in foundational local-language reading, English, and mathematics, but those programme percentages should not be mistaken for a per-turn grading algorithm. ^[15]

Lesson for Sabi: the output of grading must change the next problem, scaffold, or level. Collecting a score without changing instruction is not TaRL.

2.3 USAID / RTI EGRA: a profile of reading subskills

The Early Grade Reading Assessment (EGRA) separates components such as phonological awareness, letter-sound knowledge, familiar and invented-word reading, oral reading fluency, and comprehension. Items are coded correct, incorrect, or missing/not attempted. Timed subtasks are reported as a rate, while accuracy and comprehension remain separately interpretable. ^[8]

This prevents a common error: a child who reads quickly but inaccurately should not look equivalent to a child who reads accurately but slowly, and neither should be assumed to comprehend the text. Benchmarks and acceptable responses must be adapted to the language and orthography.

Lesson for Sabi: store subskill evidence, not just “literacy correct.” Do not collapse phoneme accuracy, word reading, oral-reading rate, and comprehension into one live-call score.

2.4 RTI EGMA: diagnose the structure of mathematical learning

The Early Grade Mathematics Assessment (EGMA) uses oral, one-to-one tasks so weak reading or writing does not obscure mathematical knowledge. Core areas include number identification, magnitude comparison, missing-number patterns, addition and subtraction at more than one level, and word problems. Some subtasks are timed; others use total correct and stopping rules. ^[9]

The EGMA toolkit warns against using the core instrument as a high-stakes test, teacher evaluation, direct report-card input, or unlinked cross-country comparison. Its strength is a profile of foundational subskills for programme diagnosis and evaluation.

Lesson for Sabi: “got the answer right” is necessary but not always sufficient for diagnosis. Store the skill and item form—fact recall, quantity comparison, operation, pattern, or word problem—so the system knows what the learner has actually demonstrated.

2.5 UNICEF: different frequencies for different decisions

UNICEF’s formative-assessment guidance proposes a practical cadence: ^[6]

Assessment type	Main purpose	Typical timing	Learners
Screening	Identify possible needs and starting levels	Beginning of year/term	All
Daily mastery monitoring	Check content taught that day and reteach immediately	Daily, inside instruction	All
Progress monitoring	Check whether below-goal learners are catching up	Weekly or every 2–4 weeks	Learners below goals
Benchmark	Compare performance with target level	Mid and/or end of year	All

UNICEF emphasizes reliability, validity for the intended purpose, alignment to skill progressions, and clear rules for using results. Assessment tasks should focus on recently taught skills when the purpose is daily mastery.

Lesson for Sabi: do not re-run a long placement test every call. Use tiny embedded checks daily and a separate periodic level check.

2.6 UNICEF MICS, UNESCO UIS, and Nigeria NALABE: external proficiency reporting

MICS directly assesses children aged 7–14 in households, including those outside school. Its strict all-components rule gives a comparable population indicator. ^[7]

UNESCO’s Global Proficiency Framework describes content and performance at “partially meets,” “meets,” and “exceeds” levels in reading and mathematics. For Grade 2 reading, meeting proficiency includes accurate connected-text reading with few errors and retrieval of explicit information. For mathematics, descriptors are organized by content domains and grade-level skills. ^[13]

Nigeria’s 2022 NALABE, published in 2025, assessed nationally representative samples at P3, P5, and JS2 and placed performance in four proficiency bands using standard-setting judgments. ^[12]

Lesson for Sabi: use these frameworks to align external claims and endline reporting. Do not directly copy their cut scores into a phone tutor’s daily mastery engine.

2.7 Save the Children and Room to Read: multi-domain, language-specific literacy

Save the Children’s Literacy Boost assesses letter knowledge, phonemic awareness, fluency, vocabulary, and comprehension, then uses the profile to inform teaching and community activities. ^[16]

Room to Read uses oral reading fluency and comprehension but cautions against importing words-per-minute benchmarks from a different language. It has worked with RTI to establish language-specific benchmarks because word length, orthography, and linguistic structure affect reading rate. ^[17]

Lesson for Sabi: Nigerian English, Pidgin, Hausa, Yoruba, Igbo, and other languages cannot safely share one pronunciation list or fluency threshold without local validation.

2.8 Mastery-learning and digital platforms: high bar, reversible states

The Education Endowment Foundation’s evidence review describes mastery learning as keeping outcomes constant while varying the time and support needed. A high mastery bar—usually 80–90% on the relevant assessment—is associated with more successful approaches, although the evidence strength is rated low and implementation quality matters. ^[18]

Khan Academy’s digital mastery system uses skill states rather than only percentages: attempted, familiar, proficient, and mastered. Evidence from later quizzes and tests can move a skill up or down. ^[19]

Lesson for Sabi: mastery should be a revisable state backed by recent evidence, not a permanent badge earned from one lucky answer.

3. Cross-organization comparison

Organization / model	Primary purpose	What is recorded	How results change teaching	Important limitation
ASER / Pratham	Rapid placement and population status	Highest reading/arithmetic level	Makes below-grade learning visible; informs grouping	Too few items for fine-grained mastery
TaRL / J-PAL	Instructional placement and regrouping	Current level and movement	Groups by level; targets activities; reassesses	Success depends on implementation system, not test alone

Organization / model	Primary purpose	What is recorded	How results change teaching	Important limitation
EGRA	Reading diagnosis/evaluation	Separate subtask accuracy, rate, comprehension	Reveals specific literacy bottlenecks	Language adaptation and trained administration required
EGMA	Mathematics diagnosis/evaluation	Separate number, operation, pattern, problem-solving subtasks	Identifies conceptual/procedural gaps	Not designed as a report-card or high-stakes grade
UNICEF formative guidance	Assessment for learning	Screening, daily, progress, benchmark evidence	Immediate feedback, reteaching, regrouping	Requires explicit rules and teacher/system capacity
UNICEF MICS / UNESCO UIS	Population and SDG reporting	Strict proficiency classifications	Guides policy and resource decisions	Not a daily tutoring algorithm
Nigeria NALABE	National proficiency reporting	Four proficiency bands	System planning and target-setting	Cut scores are test/grade/subject specific
Save the Children / Room to Read	Literacy programme diagnosis	Multi-domain reading profile	Targets instruction and materials	Fluency thresholds cannot be copied across languages
EEF / Khan Academy	Mastery progression	Skill states backed by repeated performance	Adds support, practice, reassessment	Digital/mastery rules still require local validation

4. Proposed Sabi Grading Standard v0.1

4.1 Non-negotiable principle: score evidence, not the conversation

Sabi's voice can be warm, flexible, and conversational. The scoring layer should be deliberately boring: versioned item, expected answer, accepted forms, learner response, support level, and deterministic outcome.

The LLM may:

- explain;
- encourage;
- choose an allowed scaffold;
- acknowledge the learner's wording; and
- summarize progress.

The LLM must not:

- invent the expected answer after hearing the learner;
- change a correct numeric result because the object noun differs;
- turn uncertain audio into an incorrect academic response;
- promote or demote a learner from one unverified turn; or
- overwrite the deterministic grader.

4.2 Record five dimensions for every response

Do not force all evidence into one code. Store these dimensions separately:

Dimension	Allowed values	Meaning
Audio scorability	scorable / not scorable	Was there enough reliable learner audio to judge?
Academic correctness	correct / incorrect / indeterminate	Did the demonstrated content match the item rule?
Independence	independent / self-corrected / scaffolded / modelled	How much support preceded the answer?
Prompt level	none / technical repeat / neutral probe / conceptual cue / answer model	What Sabi did before the response
Recognition evidence	provider transcripts, consensus, confidence, raw audio	Why the system interpreted the audio as it did

Rules

- “Not scorable” is excluded from accuracy denominators and triggers a neutral retry.
- A correct answer after a conceptual cue is successful guided learning, not independent mastery.
- Repeating a prompt because the line failed does not reduce independence if no academic hint was added.
- A learner who corrects herself before feedback receives “self-corrected,” which is positive evidence but remains distinguishable from first-attempt independence.
- An answer repeated after Sabi models it is practice only; it cannot prove mastery.

4.3 Skill-state model

State	Evidence meaning	Instructional action
Not introduced	No valid evidence	Introduce when prerequisites are ready
Emerging	Can respond with substantial support or occasional independent success	Model, use concrete examples, reduce difficulty
Developing	Repeated success, but not yet stable/independent	Mix guided practice with new independent probes
Secure	Meets the independent mastery rule	Advance while using short interleaved review
Retained	Passes a later spaced probe	Reduce review frequency; keep in cumulative checks

Proposed secure rule for the pilot

A skill becomes secure when all of the following are true:

1. at least four of the last five **scorable independent probe items** are correct;
2. the evidence spans at least two separate calls;
3. at least two item forms or contexts are represented;
4. repeated identical items do not count twice; and
5. no prerequisite check in the same period shows a clear unresolved gap.

A skill becomes retained after an independent correct probe at least seven days later, or after three intervening sessions if the learner calls less predictably.

One miss does not demote a secure skill. Two independent misses on different item forms move it to “review required”; a short confirmation set determines whether it remains secure or returns to developing.

This rule deliberately combines a high bar with protection against one ASR error, one lucky guess, and same-session imitation. It should be calibrated against human scoring during the pilot.

4.4 Numeracy scoring rules

Deterministic item contract

Every numeracy item must be created before the learner answers and contain:

- item ID and grader version;
- skill and prerequisite skill;
- exact mathematical representation;
- expected value and valid equivalents;
- item form (bare fact, comparison, pattern, story problem, explanation);
- difficulty parameters;
- objects/context used; and
- misconception tags that an incorrect response may indicate.

Numeric-first grading

For a question whose expected answer is 2, all of the following are academically correct:

- “two”;
- “2”;
- “two apples”;
- “two fries”; and
- an accepted equivalent in the learner’s selected language.

The object word is ignored unless the item explicitly assesses a unit or category. This prevents the conversational model from overturning correct mathematics.

Recognition and decision sequence

1. Run the audio/turn gate.
2. Preserve raw audio and all recognizer outputs.
3. Normalize number words, digits, and approved language variants to a numeric candidate.
4. If the independent recognizers agree on the number, compare it with the stored expected value.
5. If recognizers disagree or only low-quality evidence exists, ask one neutral confirmation, such as “Did you say eight?”
6. If still ambiguous, record not scorable—not wrong.
7. Generate the teaching response from the deterministic outcome.

ASR confidence and mathematical correctness are different variables. A high-confidence transcript can contain the wrong number; a low-confidence transcript may still produce a clear numeric consensus.

Accuracy, fluency, and reasoning remain separate

- **Accuracy:** is the value correct?
- **Automaticity:** how efficiently is a known fact retrieved? Response time is diagnostic, not a correctness rule.
- **Method/reasoning:** can the learner represent or explain the operation? Use a separate rubric and only when the lesson teaches reasoning.
- **Language comprehension:** did the learner understand the story problem? Do not infer this from bare-fact performance.

Multiplication-specific protections

- Use a deterministic deck with balanced factor ranges and controlled difficulty.
- Avoid repeating commutative pairs and products in the same short session unless repetition is intentional.
- Track whether the learner solved equal groups, repeated addition, arrays, or a recalled fact.
- After a scaffolded correct response, confirm with a different but structurally equivalent problem.
- Never let product frequency—for example repeated questions that all equal 12—masquerade as broad multiplication mastery.

4.5 Literacy scoring rules—and the phone limitation

An audio-only call cannot, by itself, prove that a child decoded print. It can assess oral language and phonological skills, but visual reading tasks require the child to see a letter, word, or passage through a worksheet, SMS/WhatsApp image, app screen, or other shared stimulus.

Safe on audio-only calls

- listening comprehension;
- oral vocabulary;
- syllable and phoneme blending;
- phoneme segmentation;
- identifying initial/final sounds; and
- oral manipulation tasks.

Requires a controlled visual stimulus

- letter recognition;
- grapheme–phoneme correspondence from print;
- word and nonword decoding;
- connected-text reading accuracy;
- oral reading fluency; and
- reading comprehension rather than listening comprehension.

When visual reading is added:

- validate accepted pronunciation variants with Nigerian language and literacy experts;
- score correct words, errors, and self-corrections explicitly;
- report correct words per minute and accuracy separately;
- score comprehension separately from oral-reading rate;

- keep raw audio for audit; and
- do not rely on general-purpose STT alone for isolated phonemes.

Because telephone bandwidth compresses speech and Nigerian child-voice recognition remains under-validated, literacy should remain outside consequential automated grading until human comparison shows acceptable reliability across accents, languages, devices, line conditions, and background noise.

4.6 Assessment cadence for Sabi

Cadence	Sabi activity	Decision
First 1–2 calls	Short adaptive placement probes	Starting level and prerequisite gaps
Every teaching turn	One low-stakes response record	Praise, scaffold, reteach, or continue
Every 4–6 practice items	Unscaffolded probe with a fresh item	Update developing/secure evidence
Across each week or 2–4 weeks for below-goal learners	Short progress set	Is the learner progressing; should difficulty or support change?
Roughly monthly or after a defined call block	ASER/TaRL-style level check	Confirm or change placement
Pilot baseline and endline	Standardized, human-administered or human-verified measure	Programme impact; external reporting

The benchmark instrument should not reuse the exact tutoring questions. Otherwise, familiarity with the item can be mistaken for transferable learning.

4.7 Backend event model

Every scored response should preserve:

- learner, call, session, turn, and item IDs;
- domain, skill, subskill, and prerequisite;
- item text/structure and expected answer;
- raw audio reference;
- every ASR transcript, normalized candidate, latency, and confidence;
- audio-scorability decision and reason;
- correctness, independence, prompt level, and self-correction;
- misconception or error tag;
- grader name/version and curriculum version;
- mastery state before and after the turn; and
- next instructional action.

This creates an auditable chain from “what the learner said” to “why Sabi advanced.” It also permits rescored old turns when the recognizer or grader improves, without rewriting the historical raw evidence.

4.8 Dashboard design

The owner/teacher view should prioritize decisions rather than transcript volume:

1. current TaRL-style literacy and numeracy placement;
2. skill map: not introduced, emerging, developing, secure, retained;
3. independent accuracy and supported accuracy shown separately;
4. most recent valid evidence and next review date;
5. not-scorable rate and ASR disagreement rate;
6. recurring misconception tags;
7. level movement over time; and
8. a drill-down audit trail with audio, transcripts, expected answer, support, and grader decision.

Do not show a single “82% smart” score. If a summary is needed, show the proportion of target skills secure and retained, alongside coverage and evidence quality.

5. Pilot validation plan

5.1 Human comparison is the launch gate

Before automated scores drive placement or impact claims:

1. create a Nigerian phone-audio test set covering age, gender, accent, language, device, network quality, and background noise;
2. have trained human scorers independently code the raw audio using the same response dimensions;
3. adjudicate disagreements and preserve the gold label;
4. compare Sabi with the human label for scorability, numeric extraction, correctness, and support level;
5. report false-pass and false-fail rates by subgroup and acoustic condition; and
6. adjust rules and accepted forms before expanding consequences.

5.2 Reliability checks

- **Inter-rater reliability:** do trained humans agree on the rubric?
- **Machine-human agreement:** does Sabi match the adjudicated score?
- **Test-retest stability:** does the same skill classification remain reasonably stable on equivalent items?
- **Internal evidence consistency:** do multiple items support the same mastery state?
- **Fairness:** do error rates differ materially by accent, language, sex, age, device, or network condition?
- **Decision validity:** do learners promoted by the rule succeed on a fresh transfer item?

5.3 Separate product analytics from learning impact

Operational metrics—call completion, latency, interruption success, ASR agreement, and not-scorable rate—tell us whether the phone agent works. Learning metrics—independent skill mastery, retention,

level movement, and standardized baseline-to-endline change—tell us whether children learn. Both matter; neither can substitute for the other.

6. Recommended implementation order

Phase 1: numeracy evidence integrity

- versioned deterministic item bank;
- numeric-first grader and language normalizer;
- response-dimension event schema;
- not-scorable handling;
- no repeated-product bug; and
- call-review audit trail.

Phase 2: mastery and placement

- five-state skill model;
- four-of-five independent secure rule;
- spaced retention probes;
- prerequisite graph;
- short adaptive placement and periodic level checks; and
- teacher/owner dashboard.

Phase 3: pilot measurement

- human-scored calibration set;
- baseline and endline instrument separate from tutoring items;
- subgroup reliability/fairness analysis; and
- threshold revision from evidence.

Phase 4: literacy, only after measurement validity is solved

- decide the visual channel;
- define language-specific accepted responses;
- validate phoneme/word scoring on Nigerian child phone audio;
- separate decoding, fluency, and comprehension; and
- launch first as human-review-assisted, not fully consequential automation.

7. What is evidence-backed versus proposed for Sabi

Recommendation	Status
Progressive, one-to-one placement and level-based teaching	Directly supported by ASER/TaRL/J-PAL
Separate literacy and numeracy subskills	Directly supported by EGRA/EGMA and literacy programmes

Recommendation	Status
Daily formative checks plus periodic progress/benchmark assessment	Directly supported by UNICEF and World Bank guidance
High mastery bar before advancement	Supported in mastery-learning evidence; exact implementation varies
Four of five independent items across two calls	Proposed Sabi rule; informed by ASER and 80–90% mastery practice, requires pilot validation
Prompted success does not equal independent mastery	Assessment-design inference consistent with scaffolded instruction; validate rubric use
Not-scorable audio is excluded from academic accuracy	Validity safeguard derived from distinguishing missing/unattempted from incorrect responses
Numeric value outranks irrelevant object nouns	Deterministic construct-alignment rule for Sabi
Spaced retained state	Evidence-informed product design; timing requires pilot calibration
Phone-only audio cannot prove print decoding	Construct-validity requirement

Final recommendation

Sabi should borrow the **simplicity and actionability of ASER/TaRL**, the **subskill discipline of EGRA/EGMA**, the **assessment cadence of UNICEF**, the **separation of classroom and system evidence used by the World Bank**, and the **performance-band discipline of UNESCO/NALABE**.

It should not copy any one tool wholesale. Those tools serve different decisions. The correct Sabi architecture is a linked system in which:

- conversation teaches;
- deterministic rules score;
- repeated independent evidence establishes mastery;
- progressive probes determine placement;
- standardized human-verified assessment establishes impact; and
- ambiguous phone audio never becomes a false academic failure.

For the next build, numeracy should remain the first domain. Implement the response event schema and deterministic item contract before adding more model reasoning. That foundation will make every later choice—mastery, dashboards, research reporting, and literacy validation—more trustworthy.

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All web sources were reviewed on 7 August 2026. Existing Education for Equality research on TaRL, phonics, EKOEXCEL, IVR delivery, Rori AI, and Nigerian context was reviewed before new research was conducted.